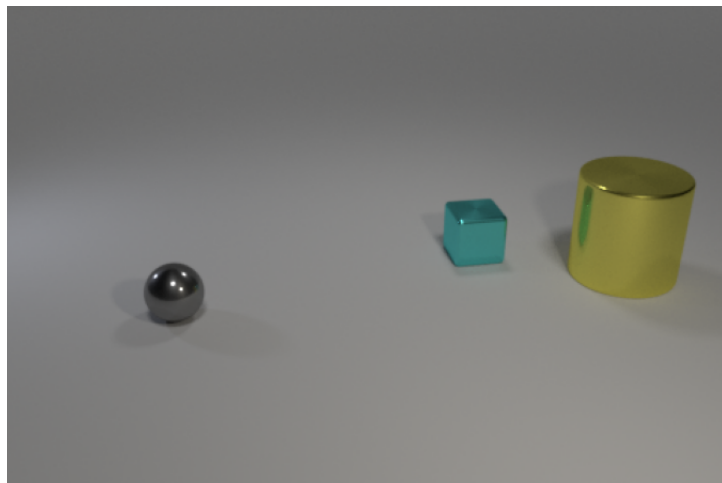


First 3 validation images, first positive/negative query each | seed 17
Shared frozen visual CNN; 2 optical layers + per-layer OEO; green=correct, red=incorrect

CLEVR_train_000951.png



is a cyan cube present in the image ?

Ground truth: YES

One-hot / MoE: YES (Pyes=0.74)

One-hot / D2NN: YES (Pyes=0.52)

GRU / MoE: YES (Pyes=0.51)

GRU / D2NN: YES (Pyes=0.73)

is a purple cylinder present in the image ?

Ground truth: NO

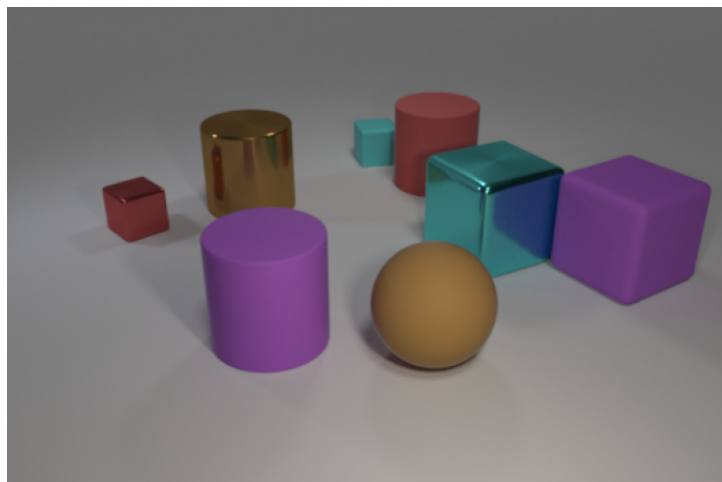
One-hot / MoE: NO (Pyes=0.29)

One-hot / D2NN: NO (Pyes=0.41)

GRU / MoE: NO (Pyes=0.00)

GRU / D2NN: NO (Pyes=0.07)

CLEVR_train_000622.png



can you see a purple cube ?

Ground truth: YES

One-hot / MoE: YES (Pyes=0.55)

One-hot / D2NN: YES (Pyes=0.52)

GRU / MoE: YES (Pyes=0.63)

GRU / D2NN: YES (Pyes=0.66)

can you see a blue cube ?

Ground truth: NO

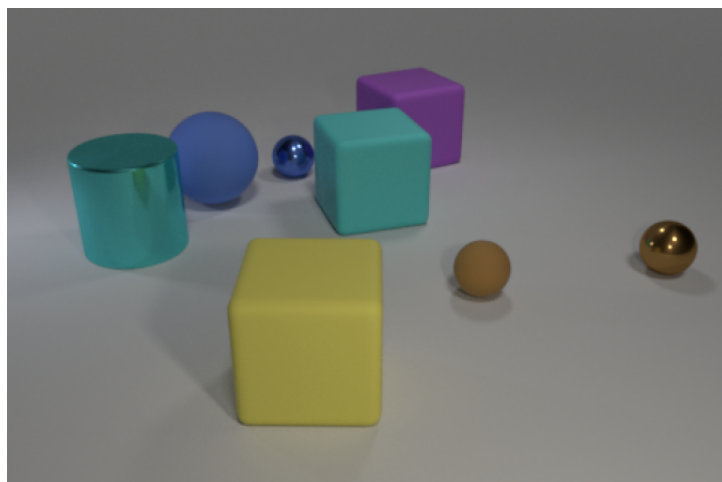
One-hot / MoE: NO (Pyes=0.12)

One-hot / D2NN: NO (Pyes=0.42)

GRU / MoE: NO (Pyes=0.16)

GRU / D2NN: NO (Pyes=0.21)

CLEVR_train_001049.png



does the image contain a brown sphere ?

Ground truth: YES

One-hot / MoE: YES (Pyes=0.87)

One-hot / D2NN: YES (Pyes=0.54)

GRU / MoE: NO (Pyes=0.43)

GRU / D2NN: YES (Pyes=0.57)

does the image contain a red cube ?

Ground truth: NO

One-hot / MoE: NO (Pyes=0.48)

One-hot / D2NN: NO (Pyes=0.48)

GRU / MoE: NO (Pyes=0.17)

GRU / D2NN: NO (Pyes=0.15)